



Fundamental Math Assessment 3: Ratios and Proportions

**** You must show all work to receive credit for each problem (when applicable)****

3.A.1.a-d Solve for x by using cross multiplication.

a. $\frac{1}{2} = \frac{x}{24}$

b. $\frac{10}{50} = \frac{x}{25}$

c. $\frac{5}{2} = \frac{x}{8}$

d. $\frac{40}{100} = \frac{8}{x}$

3.A.2 At a school board meeting, $\frac{1}{2}$ of the participants are teachers, $\frac{1}{3}$ of the participants are administrators, and the remaining 10 participants are all students. How many total participants (teachers, administrators, and students) were at the board meeting?

3.A.3.a-d Fill in the blanks (the numerators of the late fractions).

$$\frac{1}{10} = \frac{\quad}{20} = \frac{\quad}{100}$$

$$\frac{2}{3} = \frac{\quad}{6} = \frac{\quad}{90}$$

3.B.1.a-d. Convert the following fractions into percentages

a. $\frac{1}{2} = \quad \%$

b. $\frac{1}{4} = \quad \%$

c. $\frac{2}{5} = \quad \%$

d. $\frac{3}{8} = \quad \%$

3.B.2.a-d Convert the following percentages into decimals.

a. $30\% =$

b. $55\% =$

c. $120\% =$

d. $7.7\% =$

3.C.1 Solve for x if $2:7$ is the same proportion as $x:28$.

3.C.2 If $x:y$ equals $3:4$ and $y:z$ equals $4:5$, what is the proportion $x:z$ equal to in lowest terms?